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Vault Guide to the Case Interview Guesstimates

"Guesstimate" Case Interviews

The approach to guesstimates is basically the same as business cases -- you will showcase your ability to analyze a situation and form conclusions about this situation by thinking out loud. The difference here is that you will not necessarily be using a series of questions to gather feedback from the interviewer. Instead, you will drive toward a conclusion through a series of increasingly specific analyses. Let's look at an example:

How many ping-pong balls fit in a 747?

This is an actual question used in consulting interviews. If you are a little unsettled by this type of question, it's no wonder. That is exactly the reaction the interviewer is expecting. Remember that the main objective of these questions is to evaluate your poise and professionalism when facing an outlandish situation. How you react to this question when presented will speak volumes about your ability to be professional when faced with a similar business situation at a client.

So, how do you approach a guesstimate question? First, don't panic. If you are visibly shaken when presented with a guesstimate or brainteaser, it will hurt you. It is extremely important that you do not lose your cool.

Do not let yourself struggle verbally. You are free to say something like, "That is an intriguing question. May I have a moment to think it through?" This statement immediately shows the interviewer you are still in control and gives you some breathing time to think about a method for answering,

Once you have had a minute to compose your thoughts, be sure and go through your reasoning out loud, so your interviewer can see that you're arriving at your answer in a logical manner, "Don't be anal," suggests one former consultant. "You should realize that for the purposes of a guesstimate, 1,000,553 is the same as a million, and you can divide by 350 if you need to divide by the number of days in the year."

Finally, remember that there is no right answer for guesstimates. It will often not even be necessary to come up with a definitive response like "1,400,350," due to constraints on time. Always work toward a final answer, but do not

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feel that you have done a poor job if the interviewer moves on to other topics before you are finished. They may simply recognize that you're on the right track and see no reason to keep going.

Acing Guesstimates

The best approach for a guesstimate or brainteaser question is to think of a funnel. You begin by thinking broadly, then slowly drill down towards the answer. Let's look at this approach in context. Referring to our sample question, you know that you are looking for how many ping-pong balls fit in a 747 airplane. The first thing you need to determine is the volume of the ping pong ball.

For any guesstimate or brainteaser question you will need to understand whether your interviewer will be providing any direction or whether you will have to make assumptions. Therefore, begin the analysis of a guesstimate or brainteaser question with a question to your interviewer, such as, "What is the volume of a single ping pong ball?" If the interviewer does not know or refuses to provide any answer, then you will know that you must assume the answer. If the interviewer does provide the information, then your approach will be a series of questions. For this example let's assume your interviewer wants you to make the assumptions. Your verbal dialogue might go something like this:

Let's assume that the volume of a ping-pong ball is three cubic inches. Now let's assume that all the seats in the plane are removed. We'll say the average person is six feet high, one foot wide and one foot deep. That's 6 cubic feet, or 1,728 cubic inches, (One cubic foot is 12x12x12 inches, or 1,728 cubic inches.)

Okay, so a 747 has about 400 seats in it, excluding the galleys, lavatories, and aisles on the lower deck and about 25 seats on the upper deck. Let's assume there are three galleys, 14 lavatories, and three aisles (two on the lower deck and one on the upper deck), and that the space occupied by the galleys is a six-person equivalent, by the lavatories is a two-person equivalent, and the aisles is a 50-person equivalent on the lower deck and a 20-person equivalent on the upper deck. That's an additional 18, 28, and 120 person-equivalents for the remaining space. We

won't include the cockpit since someone **has** to fly the plane. So there are **about** 600 person-equivalents available. (You would be rounding a bit to make your life easier, since the actual number is 591 person equivalents.)

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In addition to the human volume, we have to take into account all the cargo and extra space -- the belly holds, the overhead luggage compartments, and the space over the passengers' head. Let's assume the plane holds four times the amount of extra space as it does people, so that would mean extra space is 2,400 person-equivalents in volume. (Obviously, this assumption is the most important factor in this guesstimate. Remember that it's not important that this assumption be correct, just that you know the assumption should be made.)

Therefore, in total we have 3,000 (or $600 + 2,400$) person-equivalents in volume available. Three thousand x 10,368 cubic inches means we have 31,104,000 cubic inches of space available. At three cubic inches per ball, a 747 could hold 10,368,000 balls. However, spheres do not fit perfectly together. Eliminate a certain percentage -- spheres cover only about 70 percent of a cube when packed -- and cut your answer to 7,257,600 balls.

You might be wondering how you would calculate all these numbers in your head! No one expects you to be a human calculator, so you should be writing down these numbers as you develop them. Then you can do the math on paper, in front of the interviewer, which will further demonstrate your analytical abilities.

You choose the numbers, so pick nice round numbers that are easy for you to manipulate. Even if you just read a study that states that there are 295 million inhabitants in the United States, no interviewer will flinch if you estimate the number of American inhabitants as 300 million.

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Quick guesstimate numbers

You'll need to grab numbers for guesstimates quickly, Here are some basic stats,

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There are approximately 295 million people in the United States (but you can round up to 300 million for the purposes of guesstimates). By comparison, there are 1.2 billion people in China.

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- There are about 100 million households in the United States.
- There are over 90 million people online in the United States.
- Approximately 50 percent of all Web users are English speakers.

Forrester Research estimates that by 2004, \$6.8 trillion will be spent on e-commerce online,

There are approximately eight million people in New York City, 13 million people in Shanghai; 3.5 million people in Los Angeles and seven million in London, England.

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The extra step

Don't forget to add the "extra step" into your guesstimate. If you're trying to figure out how many blocks there are in New York City, remember to eliminate blocks covered by Central Park (and other parks). If you're determining the number of black cars in the United States, once you've estimated the number of cars in America, make sure you estimate what percentage of them are black.

In-case guesstimates

Not all guesstimates are stand-alone questions. Many are contained within case questions, mostly in the form of a market sizing. (And what would you say is the size of the market for pork rinds in the United States?) The key here is to derive a reasonable and easily manipulated figure for your calculations.

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Practice Guesstimate Questions

The following are examples of guesstimates and suggested solutions. There is more variation here than in business cases. Whereas in business cases we ask that you make no assumptions, here you may have to make assumptions if your interviewer does not provide any additional information.

Remember to ask your interviewer each time you begin a guesstimate answer if they will provide information or if you will need to make assumptions. If you do, be sure to be perfectly clear where and when you are making assumptions

This set of questions can be presented as either assumptive or with feedback from the interviewer.

1. How many gallons of white house paint are sold in the U.S. every year?

The "start big" approach: If you're not sure where to begin, start with the basic assumption that there are 270 million people in the U.S. (or 25 million businesses, depending on the question). If there are 270 million people in the United States, perhaps half of them live in houses (or 135 million people). The average family size is about three people, so there would be 45 million houses in the United States. Let's add another 10 percent to that for second houses and houses used for other purposes besides residential. So there are about 50 million houses.

If houses are painted every 10 years, on average (notice how we deftly make that number easy to work with), then there are 5 million houses painted every year. Assuming that one gallon of paint covers 100 square feet of wall and that the average house has 2,000 square feet of wall to cover, then each house needs 20 gallons of paint. So 100 million gallons of paint are sold per year (5 million houses x 20 gallons). (Note: If you want to be fancy, you can ask your interviewer whether you should include inner walls as well!) If 80 percent of all houses are white, then 80 million gallons of white house paint are sold each year. (Don't forget that last step!)

The "start small" approach: You could also start small, and take a town of 27,000 (about 1/10,000 of the population). If you use the same assumption that half the town lives in houses in groups of three, then there are 4,500 houses, plus another 10 percent, then there are really 5,000 houses to worry about. Painted every 10 years, 500 houses are being painted in any given year. If each house has 2,000 square feet of wall and each gallon of paint covers 100 square feet, then each house needs 20 gallons - and so 10,000

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HELPFUL HINTS AND TIPS

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gallons of house paint are sold each year in your typical town. Perhaps 8,000 of those are white. Multiply by 10,000 - you have 80 million gallons,

Your interviewer may then ask you how you would actually get that number, on the job, if necessary. Use your creativity - contacting major paint producers would be smart, putting in a call to HUD's statistics arm could help, or even conducting a small sample of the second calculation in a few representative towns is possible.

2. What is the size of the market for disposable diapers in China?

Here's a good example of a market sizing. First, you might estimate that there are one billion people living in China. Because the population of China is young, a full 600 million of those inhabitants might be of child-bearing age. Half are women, so there are about 300 million Chinese women of childbearing age. Now, the average family size in China is restricted, so it might be 1.5 children, on average, per family. Let's say two-thirds of Chinese women have children. That means that there are about 300 million children in China, How many of those kids are under the age of two? About a tenth, or 30 million. So there are at least 30 million possible consumers of disposable diapers.

To summarize:

1 billion people

60% childbearing age

600,000,000 people

1/2 are women

300,000,000 women of childbearing age

2/3 have children

200,000,000 women with children

1.5 children each

300,000,000 children 1/10 under age 2

30 million children

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3. How many square feet of pizza are eaten in the United States each month?

Take your figure of 300 million people in America. How many people eat pizza? Let's say 200 million. Now let's say the average pizza-eating person eats pizza twice a month, and eats two slices at a time. That's four slices a month. If the average slice of pizza is perhaps six inches at the base and 10 inches long, then the slice is 30 square inches of pizza. So four pizza slices would be 120 square inches. Therefore, there are a billion square feet of pizza eaten every month.

To summarize:

There are 300 million people in America.

Perhaps 200 million eat pizza.

The average slice of pizza is six inches at the base and 10 inches long = 30 square inches (height x half the base),

The average American eats four slices of pizza a month.

Four pieces x 30 square inches = 120 square inches (one square foot is 144 inches), so let's assume one square foot per person,

Your total: 200 million square feet a month.

4. How many pay phones are there on the island of Manhattan?

There are two ways to handle this problem. First of all, you could estimate how many blocks there are in Manhattan, and assume that 75 percent of all blocks have a pay phone. (Remember, the interviewer didn't say they had to work.)

If there are about 15 avenues across Manhattan, and if the island is 300 streets long, then there are about 4,500 intersections. If every intersection indicates a block, then there are 4,500 blocks in the city. That means 3,000 pay phones. Now add the extra step and subtract the size of Central Park. Say that Central Park is 50 streets by six avenues - that means you lose 300 blocks, or 200 pay phones. You have the figure of 2,800 pay phones. Now estimate how many pay phones exist in bars, restaurants, schools, etc. There may be a total of 3,500 pay phones in Manhattan. (This question is also sometimes used with estimating the number of manhole covers in Manhattan.)

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5. How many weddings are performed each day in Japan?

Try a ground-up approach. In a city of 1 million (Kyoto), how many people are of marriageable age? Let's say 750,000. How many get married in a given year? Maybe 2 percent? That's 15,000. Now, the population of Japan is about 150 million, so multiply 15,000 by 150 – and you get 2.25 million weddings every year. Divide that by 365 and you get 6,164 weddings per day (on the

average, though clearly some days are more popular than others).

To summarize:

There are 1 million people in Kyoto.

750,000 are of marriageable age.

2 percent get married in the average year.

$750,000 \times 0.02 = 15,000$ marriages every year in Kyoto.

There are 150 million people in Japan.

$150 \times 15,000 = 2.25$ million weddings per year.

$2,250,000/365 = 6,164$ weddings per day.

6. How many children are born every day in the United States?

The population of the United States is 300 million (approximately). Half are women (150 million). Perhaps half of those are of childbearing age. How do you determine how many women are pregnant at any given time? Well;

Let's say the average span of childbearing for a woman is 40 years.

The average woman has two children.

So a woman is pregnant one year in 20, or 5 percent of the time.

3.75 million women are pregnant every year.

Divide by 365 - you get about 10,000 babies a day (actually, 10,273).

Do the extra step-round up for multiple births, so perhaps 12,000.

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7. How much change would you find on the floor of an average mall?

This seemingly silly guesstimate was received by a job seeker at McKinsey, It's an example of a guesstimate that is also a way to test candidate's "out-of-the-box" thinking. First, estimate how many stores there are in the average inall - say, 50. Now, how many people enter the average store on the average day? A thousand? So if there are 50,000 visitors to a mall daily, how many lose change? If one in 50, say, drops money (1,000 people a day), how much is the average loss of change? Most amounts are probably small. People carry fewer quarters, for example, and are more likely to retrieve them. So let's say that if a person is equally likely to drop a penny, nickel, or dime, then the average person who loses change loses a nickel. That means there would be \$500 worth of change on the average floor. If half of that change has been picked up immediately, that would be \$250 worth of change.

Also ask: Is there a fountain in the mall? If a fountain is considered to be the "floor" of the mall, the amount of change would obviously increase.

8. How many bottles of wine are consumed in the United States each week?

Determine:

1. The number of people in the United States.
2. The number of adults.

3. The number of wine drinkers.
4. Average number of glasses of wine consumed per week.
5. Number of glasses of wine in an average bottle.

(Extra step: You may wish to estimate how much wine is used for non drinking purposes - cooking, for example. Also clarify whether the interviewer is speaking of standard-sized bottles of wine.)

You could reasonably make the following assumptions:

There are about 300 million people in the United States.

250 million are adults.

Perhaps 220 million drink alcohol.

200 million drink wine.

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The average wine drinker drinks two glasses of wine a week.

400 million glasses of wine consumed per week.

About five glasses of wine in the average bottle.

80 million bottles of wine consumed in America each week.

Estimate how many bottles of wine are used for cooking - perhaps another five million,

This brings your total to 85 million bottles of wine consumed per week.

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More Guesstimates...

9. How many men's suits were sold in the United States last year?

Determine:

1. Estimate the population of the United States.
2. Cut that in half to get the number of men,
3. Determine how many men are employed in occupations which require business attire.
4. Determine how many suits the average business attire employee would have purchased (due to weather, fashion changes, cleanings, wear, and so on).
5. Assume (or ask) a number of suits that those men not employed in business attire jobs purchased for religious, social, or other reasons.
6. Sum up the number and present your answer.

To summarize:

The population of the United States: 270 million (or 300 million, to round up).

Half are men - so 150 million.

Let's say two-thirds are employed.

150 million x $\frac{2}{3}$ = 100 million.

Assume every employed man owns one suit = 100 million suits.

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Estimate that about a quarter of men are in a field where they must own more than one suit.

If each of those men has an additional two suits, 25 million men $\times$ 2 additional suits = 50 million suits.

150 million suits in the United States.

How often does the average suit-owner replace a suit? Perhaps once every three years.

There are 50 million suits sold every year in the United States.

10. How many tennis balls fit in a swimming pool?

Determine:

1. What is the shape and depth of the swimming pool?
2. What is the volume of a tennis ball?
3. Estimate the volume of the swimming pool based on the depth, length, and width of the pool.
4. Calculate the number of balls by dividing the volume of the pool by the volume of the tennis ball.
5. You may wish to subtract balls due to steps in the pool and the gradual upslope of the pool due to the varying depth (but ask first).

11. How many gas stations are there in Los Angeles?

Determine:

1. What is the population of Los Angeles?
2. What is the number of cars in Los Angeles? What is the average number of cars per person (including commuters)?
3. How many gas stations are needed per car?

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12. What is the annual size of the golf ball market in the United States? What factors drive demand?

Determine:

1. What is the population of the United States? 2. What percentage of the population golfs? 3. How often does the average golfer golf?
4. What is the number of balls used in an average golf game (and number that are lost)?

What factors drive demand?

- How many golf courses are being built in the United States?
- How many are planned?
- Is the population of golfers (due to Tiger Woods and his ilk) expanding?

13. Estimate the total revenues of Disney's Tarzan .

Here's where some prior knowledge comes in handy.

Determine:

1. What were movie ticket revenues in the United States for *Tarzan*?
2. What percentage of worldwide sales is the U.S. revenue?
3. How much does Disney make from video sales?
4. What are Disney's revenues from cross licensing agreements (dolls, posters, etc.)?

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And Try These for More Practice...

14. How many red cars are there in the United States?

15. What is the annual demand for table napkins in the United States?

16. How many times would the population of China circle the globe if they held hands?

17. What is the size of the poultry market in the United States?

18. How many people are cremated in the United States every year?

19. How many barbers are there in New Orleans?

20. What is the annual market for peaches in the United States?

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